Design and Implementation of a Non-coupled Integrated Transformer for a Two-phase Interleaved Flyback PFC Rectifier

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***Abstract*—Universal serial bus power delivery (USB PD) is a widely-applied standard for electronic device charging. In 2021, USB PD 3.1 was announced with a variable output of 5 V to 48 V and a maximum output power of 240 W. A multi-phase flyback rectifier is capable of power factor correction (PFC), galvanic isolation, and wide output voltage range, and is proposed as an alternative topology for USB PD. However, the number of magnetic components increases with the number of phases, which can lead to lower power density. This paper proposes the design of a non-coupled integrated transformer for a two-phase interleaved flyback PFC rectifier to reduce the size and number of magnetic components. Independent operation of the two phases using the proposed transformer was verified through simulation and experiments using a 240-W prototype operating at 100 kHz in discontinuous conduction mode (DCM). After conducting a volume analysis through the area product (AP) method, the proposed transformer is estimated to achieve up to 16% volume reduction compared to its conventional, discrete counterpart.**

***Index Terms*—Integrated magnetics, Transformer design, Two-phase interleaved flyback rectifier, USB PD, PFC, Core volume analysis.**

I. Introduction

UNIVERSAL serial bus power delivery (USB PD) is a technology used for convenient, portable, fast charging in electronic devices. The USB PD 3.1 standard, announced in 2021, expanded the maximum supplied power up to 240 W, allowing for higher power applications [1]. Adapters for USB PD 3.1 must be capable of power factor correction (PFC) for supporting 75 W and above, be equipped with galvanic isolation for user safety, and be capable of providing an output voltage of 5 V to 48 V [2]. Furthermore, USB PD 3.1 incorporates an adjustable voltage supply (AVS) to optimize charging efficiency within the extended power range (EPR), which is designated for power delivery exceeding 100 W up to a maximum of 240 W. The recent USB PD 3.2 standard, announced in 2023, further emphasizes this capability by expanding AVS mandates to the standard power range (SPR), which encompasses applications operating at or below 100 W [3].

A flyback rectifier has been widely used for commercial adapters due to its simple configuration, easy PFC control, and wide output voltage range. Improved configurations of flyback rectifiers such as active clamp flyback converters [4], [5] and bridgeless flyback rectifiers have also been proposed for higher converter efficiency [6], [7], [8], [9]. However, they are generally more practical for lower power applications due to the inherent high current stress on the switches by these topologies.

Several power supply configurations for USB PD have been proposed to meet the requirements. One example is a combination of a PFC rectifier, an LLC converter, and a buck converter as shown in Fig. 1(a) [10], [11]. While this configuration achieves high power density, the additional buck stage required to compensate for the limited gain range of the LLC increases the total component count and system cost. An asymmetrical half bridge (AHB) flyback converter in the second stage of a boost PFC circuit was introduced as shown in Fig. 1(b) [12]. The AHB allows for zero-voltage switching (ZVS) by the application of resonant capacitors in series placed on the primary side of the transformer. As a single-phase topology, the AHB processes the entire load power through a single transformer and switch pair. While this minimizes the component count, it inherently concentrates thermal stress in specific circuit elements.

Interleaving rectifiers have been considered to increase the maximum output power. Two-phase interleaving allows half of the total power to be supplied in each phase, therefore reducing the possible stress on the circuit elements [13]. The topology is shown in Fig. 1(c), where the two phases are parallel in the input and output side. The source nodes of switches S_1 and S_2 are common, and only one power supply is required for gate drive circuits. A common cathode package can be used for diodes D_1 and D_2, therefore fewer components are used for the circuit.

Operating multi-phase flyback converters across a wide output range presents some challenges. For instance, to ensure high efficiency across wide-range applications, converters must commonly employ phase-shedding, for which various control schemes have been proposed [14], [15], [16]. This

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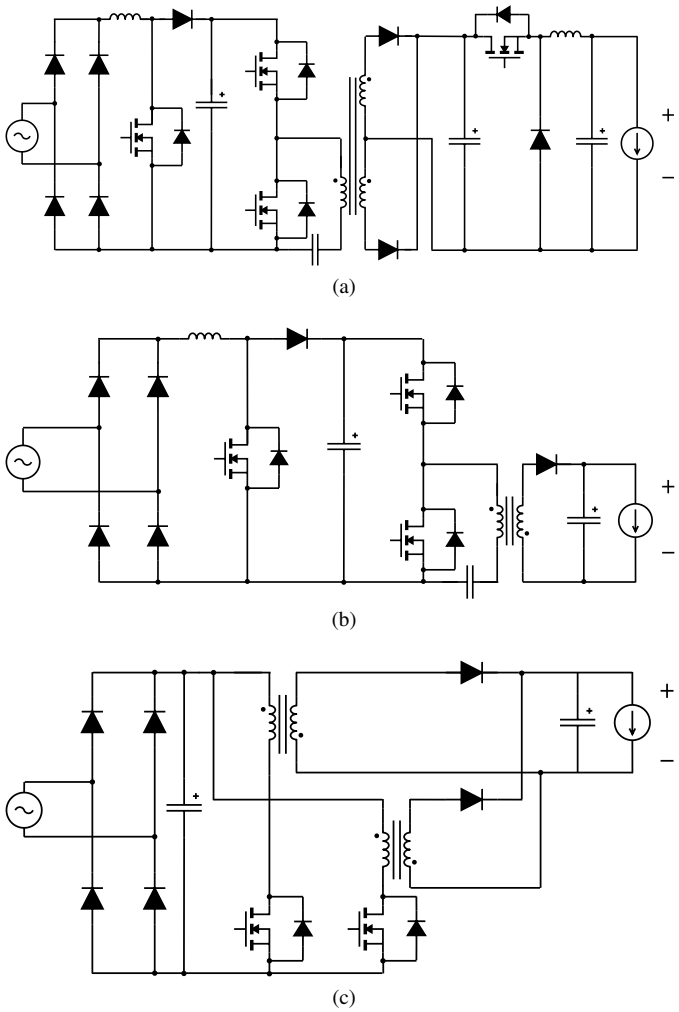


Fig. 1. Power supply configurations proposed for PD applications: (a) PFC boost, LLC, and buck converter combination in [10], [11], (b) PFC boost with AHB flyback [1], (c) two-phase interleaved flyback PFC rectifier.

strategy deactivates specific operational phases depending on the load condition to optimize light-load efficiency. However, the implementation of these control schemes often necessitates complicated designs and additional active components, thereby increasing the circuit complexity.

Moreover, increasing the number of phases inherently means an increase in the magnetic components required for the circuit. In the topology utilized for this study, although fewer active components are used overall, two discrete transformers are required, which compromises the power density of the converter. Integrated magnetics have been proposed and widely studied to improve the power density of various types of converters. Because integrated magnetics consolidate multiple discrete components through separate windings on a single magnetic core, these components inherently share the same flux paths, which translates to magnetic coupling. Consequently, the winding structure requires careful design to control magnetic flux.

While magnetic coupling offers certain advantages, it introduces operational limitations. When mutual coupling exists between windings, the phases influence one another. In some coupled topologies, utilizing phase-shedding for single-phase operation at light loads induces a voltage across the inactive phase. This generates circulating currents that degrade light-load efficiency [17], [18]. Even under heavy-load conditions with multiple active phases, coupling can facilitate power delivery across non-operational phase intervals, leading to input current distortion and additional induced losses, therefore reducing overall efficiency [17]. Moreover, sharing flux paths within a single core can introduce severe parasitic challenges for flybacks. The physical proximity of shared winding due to tight coupling in some integrated transformers introduces severe inter-winding stray capacitances, which can trigger high-frequency resonances and current spikes during switching transitions, and significantly increase power losses [19]. Some integrated planar transformers for flybacks struggle to maintain zero-voltage switching (ZVS) across wide input ranges due to the severe parasitic capacitances inherent to their multi-layer PCB structures [20]. With coupling, achieving optimal operation under asymmetrical conditions must be more difficult with these added parasitics, and can thus affect reliability.

For heavy-load applications, however, magnetic coupling can be beneficial. Coupling passively forces current equalization and minimizes capacitive current ripple, while the integrated structure improves power density. This intentional coupling is utilized in fixed-load server power supplies and LLC resonant converters [21], [22], [23]. Furthermore, modern interleaved flybacks require additional active switching devices for clamp circuits to ensure this current sharing [24]. Ultimately, while coupled magnetic structures are effective for fixed heavy-load targets, they conflict with the independent phase operation required by wide-range, variable-power converters.

Recent developments in interleaved flyback rectifiers have focused heavily on optimizing switching performance and power density. Several recent interleaved topologies still utilize discrete transformers to ensure simple control and independent operation [25], [26], [27], [28], [29]. To offset the volumetric penalty of using discrete cores, these designs often incorporate additional auxiliary components to realize distinct operational benefits. For example, discrete multi-transformer configurations have been paired with auxiliary branches to enable active clamp soft-switching [25], while other designs utilize clamping diodes or auxiliary switches to recycle transformer leakage energy [26], [27]. Furthermore, adopting wide-bandgap GaN power devices allows these discrete configurations to operate at ultra-high switching frequencies up to 1 MHz [28]. Additionally, an approach proposed in [29] enables wide output voltage gain and bidirectional capability by introducing an additional coupled inductor at the output stage. While these discrete approaches effectively protect switching devices, improve conversion efficiency, and extend operational capabilities, they inherently compromise overall system power density and increase footprint by requiring multiple separate magnetic cores.

To overcome the volume limitations of discrete implementations, integrated magnetic structures for interleaved flybacks have also been researched [30], [31], [32]. In these configurations, magnetic coupling is intentionally introduced to realize distinct operational benefits. For instance, intentional coupling

between phases of the integrated transformer using an EE-core has been utilized to achieve reduction of ripple current [30]. Coupling between four windings of a coupled inductor was utilized to enable soft-commutation in current-fed PFC rectifiers [31]. However, these approaches require complex tuning of the coupling coefficients and design parameters. The transformer in [32] utilizes an EE-core with inversely coupled secondary windings to meet specific output current ripple requirements. However, the inverse coupling may introduce unwanted circulating currents or EMI if not properly managed. Therefore, the introduction of coupling can introduce some complications in manufacturing. For interleaving, coupling makes control design more complicated, as the coupling coefficients have to be taken into account in the transfer functions, since coupling causes cross talk between the phases [33].

Structural decoupling approaches have been explored to bypass these circulating currents and accommodate the variable-load requirements of USB PD, offering an alternative to control-based phase shedding. Some approaches design decouples phases by separating the flux paths using 3-D orthogonal-flux core geometries [34], [35]. However, these custom cores constrain commercial manufacturability and increases fabrication costs. Another proposed solution is an integrated nonlinear coupled inductor designed to decouple four phases at light loads [18]. Achieving this variable coupling requires customized, multi-piece L- and T-shaped core assemblies with manually configured geometries, introducing tight assembly tolerances and manufacturing complexity.

This paper proposes a non-coupled integrated transformer for a two-phase interleaved flyback rectifier as shown in Fig. 2(a) to address these limitations. The structure is designed to achieve zero magnetic coupling between phases, ensuring independent phase operation without magnetic interference. The proposed winding structure is shown in Fig. 2(b) where N_{ab} is the number of turns between terminals a and b. Half of N_{12} and N_{34} are wound at each outer leg with opposite polarity, so as to allow for flux cancellation at the center leg. On the other hand, N_{56} and N_{78} are also wound with opposite polarity on the center leg, and allows for the transformer N_{12} and N_{34} to not operate during its operation.

The zero-coupling coefficient between the two separate transformers simplifies implementation of rectifiers and transformer design. The operation of the rectifiers in each interleaved phase still follows that of a conventional flyback converter, preserving simplicity in control and timing. The design is also highly dependent on the duty cycle, an easily adjustable parameter in circuit design. The converter in this study aims to operate in DCM in order to allow for easier implementation of PFC and to conform to the standards of USB PD. It also realizes independent phase operation, allowing the circuit to function in one-phase mode if desired, thereby enhancing flexibility in circuit operation and allowing simple implementation of phase-shedding. Furthermore, this entirely decoupled winding structure is easily achievable using a conventional EE-core, successfully bypassing the manufacturing complexities of custom geometries.

This article is an extended research of [36]. The preliminary study in [36] focused on the foundational design and experimental validation of the non-coupled integrated transformer. Key additions include a more detailed theoretical analysis, simulation validation of the operational principle, transformer design methodology including air-gap design, and a comprehensive core volume analysis utilizing the area product (AP) method. The rest of this paper is organized as follows: Section II presents an operational analysis of a two-phase interleaved flyback rectifier with the integrated transformer, along with its design method. Section III describes the size reduction effect of the transformer. Sections IV and V discuss the results of the simulation and experimental verification respectively, and Section VI presents the conclusions of this paper.

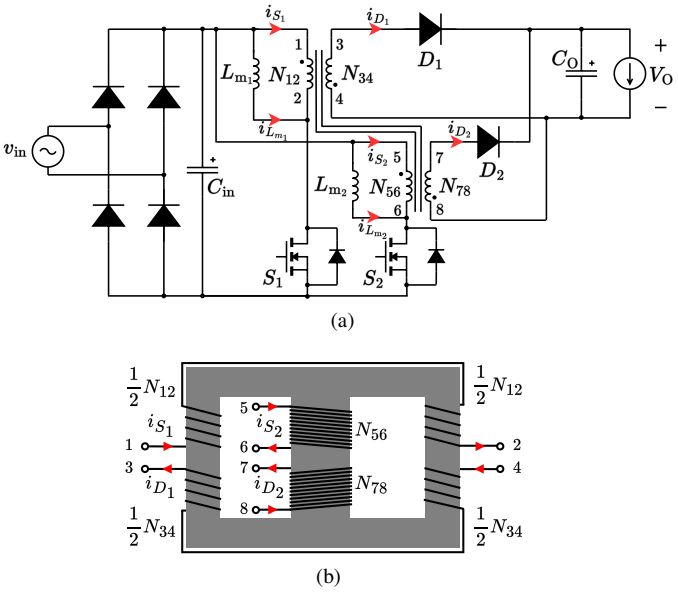


Fig. 2. The two-phase interleaved flyback rectifier (a), with the transformer and its winding structure in (b).

II. Design Method of the Proposed Transformer

A. Operation Analysis of the Proposed Flyback Rectifier and Transformer

The proposed integrated transformer, shown in Fig. 2(b) was designed for a two-phase interleaved flyback PFC rectifier, shown in Fig. 2(a) operating in discontinuous conduction mode (DCM). The switches (S_1, S_2 in Fig. 2(a)) operate with a fixed duty cycle D, fixed switching frequency f_s, and a 180-degree phase shift. The input capacitor C_{in} acts as a high-frequency filter to bypass the switching current ripple. Because it is connected directly after the bridge diode, the voltage across C_{in} inherently follows the rectified AC line voltage. Therefore, its required voltage rating must be selected to safely exceed the peak AC input voltage.

Half of N_{12} and N_{34} are wound at each outer leg with opposite polarity. Moreover, N_{56} and N_{78} are also wound with opposite polarity at the center leg. It should be noted that Fig. 2(b) is a simplified representation of the winding structure, so that the location and direction of each winding can be easily pinpointed. The transformer utilizes an EE core with a center leg that has a cross-sectional area twice its outer leg, in accordance with manufacturing specifications.

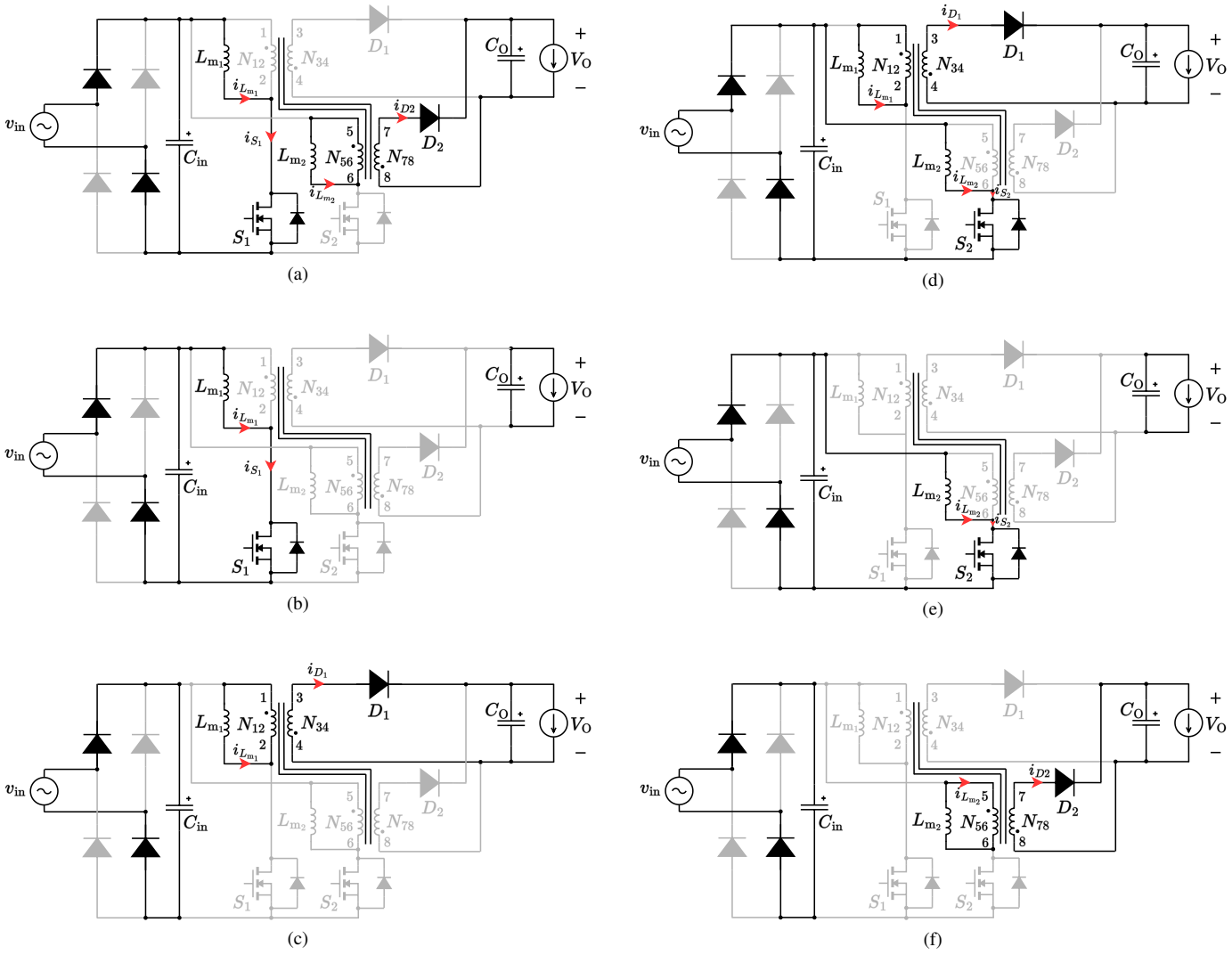


Fig. 3. Operational principle of the two-phased interleaved flyback rectifier with the proposed integrated transformer at $D \leq 0.5\frac{nV_\mathrm{o}}{V_{\mathrm{in}_\mathrm{max}}}$. There are six distinct operational modes shown: (a) Mode 1 ($t_1 < t < t_2$), (b) Mode 2 ($t_2 < t < t_3$), (c) Mode 3 ($t_3 < t < t_4$), (d) Mode 4 ($t_4 < t < t_5$), (e) Mode 5 ($t_5 < t < t_6$), and (f) Mode 6 ($t_6 < t < t_7$).

The magnetizing inductances for both transformers $N_{12}-N_{34}$ and $N_{56}-N_{78}$ were set to be equal, and are denoted as L_{m_1} and L_{m_2} respectively. The saturation magnetic flux in each separate transformer during each phase is expected to be consistent.

The operational principle of the rectifier with the proposed transformer is illustrated in Fig. 3. Additionally, Fig. 4 presents the key waveforms of the six distinct operation modes. The operation modes are examined under six-time intervals relative to the switching period T_s of S_1. The rectifier is designed with $D \leq 0.5\frac{nV_\mathrm{o}}{V_{\mathrm{in}_\mathrm{max}}}$ with n as the turns ratio, V_o as the output voltage, and $V_{\mathrm{in}_\mathrm{max}}$ as the peak value of the input voltage. This condition ensures the magnetizing current of one phase returns to zero before the other phase peaks. This duration is defined as the discharge time t_dis and is given in terms of circuit parameters by (1):

$$t_\mathrm{dis} = \frac{V_{\mathrm{in}_\mathrm{max}} D T_\mathrm{s}}{nV_\mathrm{o}} \tag{1}$$

with T_s as the switching period. For the condition given by D in this analysis, t_dis corresponds to the interval between t_3 and t_5 in Fig. 4.

To describe the magnetic state of the core, the superposition principle is used, as shown in Fig. 5(a). The magnetic flux density component B' is generated by the transformer active in the first phase defined as $N_{12}-N_{34}$. The magnetic flux density component B'' represents the component generated by the transformer active in the second phase defined as $N_{56}-N_{78}$. The total flux in the core legs is the algebraic sum of these components. This superposition determines the peak flux density required for the core design, which is analyzed in detail in Section II-B.

Modes 1, 2, and 3 in Figs. 3(a)-(c) cover the operation during the first active phase, while Modes 4, 5, and 6 in Figs. 3(d)-(f) describe the operation during the second active phase. For the following modes, B'_{o_1}, B'_{o_2}, and B'_c represent the magnetic flux densities in the respective left, right, and center legs of the core caused by $N_{12}-N_{34}$, respectively, which are shown in Fig. 5(b). On the other hand, B''_{o_1}, B''_{o_2},

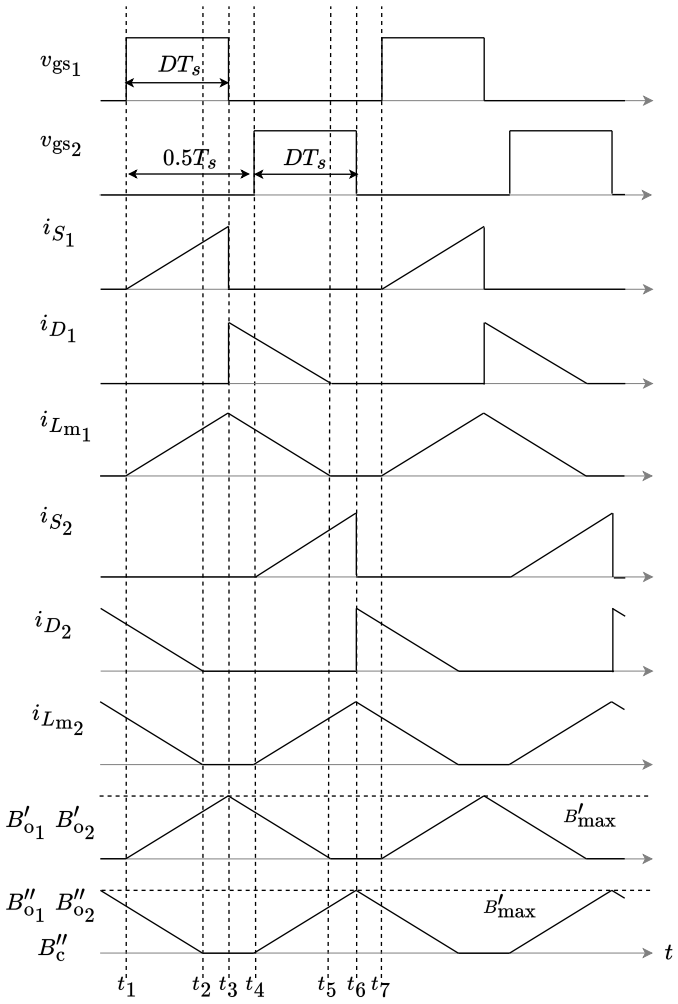


Fig. 4. Key waveforms in the operation of the two-phase interleaved flyback rectifier with the proposed transformer, with $D \leq 0.5\frac{nV_o}{V_{in_{max}}}$. S_1 and S_2 operate 180° out of phase and have the same T_s. The flux densities B' and B'' represent the individual, un-superimposed magnetic flux components generated in the first and second active phase by $N_{12}-N_{34}$ and $N_{56}-N_{78}$ respectively.

and B''_c represent the magnetic flux densities in the respective left, right, and center legs of the core caused by $N_{56}-N_{78}$, respectively, which are detailed visually in Fig. 5(c). The flux paths in the core caused by each transformer across the operational analysis illustrated in Fig. 3 are illustrated in Fig. 6.

The detailed analysis of each operating mode is as follows:

1) Mode 1 ($t_1 < t < t_2$, Fig. 3(a)): At time t_1, switch S_1 turns on and S_2 remains off. The transformer $N_{12}-N_{34}$ begins storing energy in its magnetizing inductance. Its magnetizing current $i_{L_{m_1}}$ and the corresponding flux densities in the outer legs B'_{o_1} and B'_{o_2} linearly increase from zero. Simultaneously, transformer $N_{56}-N_{78}$ completes its discharge phase, with its magnetizing current $i_{L_{m_2}}$ and secondary diode current i_{D_2} decreasing to zero at t_2. During this interval, the magnetic fluxes from both transformers coexist in the core, as illustrated in Fig. 6(a).
2) Mode 2 ($t_2 < t < t_3$, Fig. 3(b)): Mode 2 starts right after transformer $N_{56}-N_{78}$ completes its discharging phase, which means its associated currents i_{D_2}, $i_{L_{m_2}}$, and flux densities B''_{o_1}, B''_{o_2}, and B''_c are zero. Now, with S_1 still on, transformer $N_{12}-N_{34}$ continues storing energy until $i_{L_{m_1}}$ reaches its peak value at t_3.

Because the converter operates as a PFC rectifier, the instantaneous input voltage follows a rectified sinusoidal envelope $v_{in}(t) = V_{in_{max}}|\sin(2\pi f_{line}t)|$, where $V_{in_{max}}$ is the peak AC line voltage and f_{line} is the line frequency. Assuming $v_{in}(t)$ is quasi-static during a single high-frequency switching cycle, the time-variant peak primary current $i_{S_{1_{pk}}}(t)$ reached at the end of the ON-time of the first active phase is given by:

$$i_{S_{1_{pk}}}(t) = \frac{v_{in}(t)}{L_{m_1}} DT_s \tag{2}$$

where D is the duty cycle, T_s is the switching period, and L_{m_1} is the magnetizing inductance of transformer $N_{12}-N_{34}$. Consequently, the flux densities B'_{o_1} and B'_{o_2} also reach their cycle-specific peak values $B'_{o_{pk}}(t)$ which are given by (3):

$$B'_{o_{pk}}(t) = \frac{L_{m_1} i_{S_{1_{pk}}}(t)}{N_{12}A_o} \tag{3}$$

with N_{12} as the number of turns in the primary side of the active transformer and A_o is the cross-sectional area of the outer leg, where the transformer $N_{12}-N_{34}$ is wound.

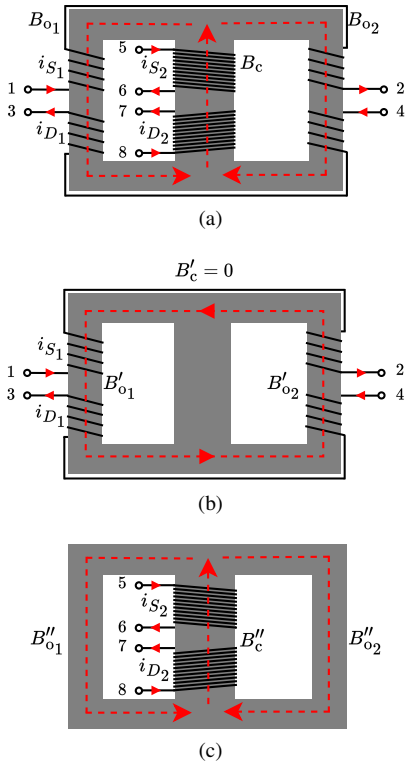


Fig. 5. The proposed integrated transformer during the two phases of the converter, with the flux density components in each leg illustrated in (a). (b) is the flux density contribution by $N_{12}-N_{34}$ during the first active phase, while (c) is the flux density contribution by $N_{56}-N_{78}$ during the second active phase.

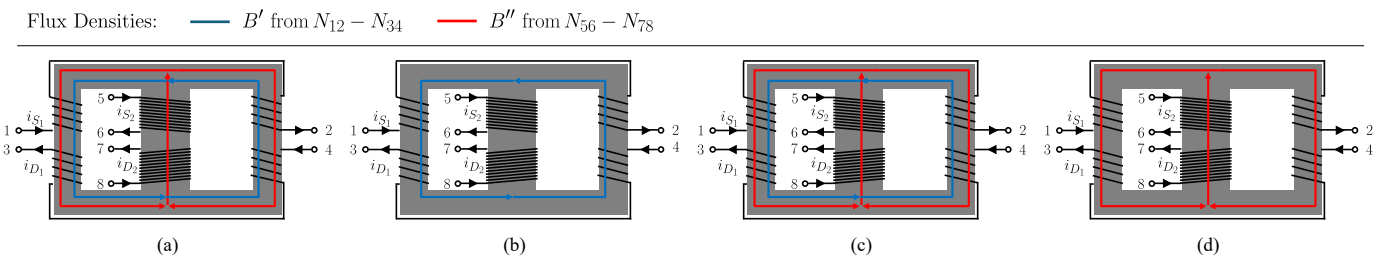


Fig. 6. Magnetic flux paths in the proposed integrated transformer across the six operating modes. (a) Mode 1 ($t_1 < t < t_2$): Phase 1 is charging and Phase 2 is discharging, resulting in the coexistence of B' and B''. (b) Modes 2 and 3 ($t_2 < t < t_4$): Phase 1 is solely active, demonstrating perfect flux cancellation in the center leg. (c) Mode 4 ($t_4 < t < t_5$): Phase 2 is charging and Phase 1 is discharging, with both B' and B'' present. (d) Modes 5 and 6 ($t_5 < t < t_7$): Phase 2 is solely active.

3) Mode 3 ($t_3 < t < t_4$, Fig. 3(c)): Mode 3 starts right after transformer $N_{12}-N_{34}$ finishes charging. At t_3, Switch S_1 turns off, initiating the discharge phase of transformer $N_{12}-N_{34}$. The magnetizing current $i_{L_{\mathrm{m}_1}}$, diode current i_{D_1}, and flux densities B'_{o_1} and B'_{o_2} begin to decrease linearly. Transformer $N_{56}-N_{78}$ remains inactive in this period.

Transformer $N_{12}-N_{34}$ operates as follows. The magnetomotive force and magnetic resistance of both outer legs for $N_{12}-N_{34}$ are equal. This causes the magnetic flux from the outer legs that flow into the center leg to be canceled. This causes no electromotive force generated in transformer $N_{56}-N_{78}$. As illustrated in Fig. 6(b), during Modes 2 and 3, the flux solely flows on the outer legs. Therefore, the transformer formed by the outer legs operates independently from the transformer in the center leg. The equivalent magnetic circuit for this transformer is shown in Fig. 7(a).

4) Mode 4 ($t_4 < t < t_5$, Fig. 3(d)): Mode 4 operates similarly to the positive cycle. Switch S_2 turns on at t_4 and S_1 remains off. Transformer $N_{56}-N_{78}$ stores energy in its magnetizing inductance. Its current $i_{L_{\mathrm{m}_2}}$ and the corresponding flux densities B''_{o_1}, B''_{c} and B''_{o_2} linearly increase. Simultaneously, transformer $N_{12}-N_{34}$ completes its discharge phase, with $i_{L_{\mathrm{m}_1}}$ and i_{D_1} decreasing to zero at t_5. Similar to Mode 1, B' and B'' coexist, as shown in Fig. 6(c).
5) Mode 5 ($t_5 < t < t_6$, Fig. 3(e)): Mode 5 starts after the discharge of transformer $N_{12}-N_{34}$ completes. Now, with Switch S_2 still on, transformer $N_{56}-N_{78}$ continues storing energy until $i_{L_{\mathrm{m}_2}}$ reaches its peak value at t_6. Similarly bounded by the input voltage envelope, the time-variant peak primary current $i_{\mathrm{S}_{2_{\mathrm{pk}}}}(t)$ reached at the end of the ON-time of the second active phase is given by:

$$i_{\mathrm{S}_{2_{\mathrm{pk}}}}(t) = \frac{v_{\mathrm{in}}(t)}{L_{\mathrm{m}_2}} D T_{\mathrm{s}} \tag{4}$$

where L_{m_2} is the magnetizing inductance of transformer $N_{56}-N_{78}$. Consequently, the flux densities B''_{o_1}, B''_{c} and B''_{o_2} also reach their cycle-specific peak values. The time-variant peak value of the center leg flux density $B''_{\mathrm{c_{pk}}}(t)$ is given by (5):

$$B''_{\mathrm{c_{pk}}}(t) = B''_{\mathrm{o_{pk}}}(t) = \frac{L_{\mathrm{m}_2} i_{\mathrm{S}_{2_{\mathrm{pk}}}}(t)}{N_{56} A_{\mathrm{c}}} \tag{5}$$

with N_{56} as the number of turns in the primary side of the active transformer and A_{c} is the cross-sectional area of the center leg where the transformer $N_{56}-N_{78}$ is wound.
6) Mode 6 ($t_6 < t < t_7$, Fig. 3(f)): Switch S_2 turns off, initiating the discharge phase of transformer $N_{56}-N_{78}$. The currents $i_{L_{\mathrm{m}_2}}, i_{\mathrm{D}_2}$, and flux densities B''_{o_1}, B''_{c}, and B''_{o_2} begin to decrease linearly. Transformer $N_{12}-N_{34}$ remains inactive.

Transformer $N_{56}-N_{78}$ operates as follows. The magnetic resistance of both outer legs is equal, causing the magnetic flux from the center legs to be divided equally into both outer legs. Since the number of turns of both outer legs is equal and wound in an opposite polarity, the electromotive force generated in $N_{12}-N_{34}$ is canceled. As illustrated in Fig. 6(d), during Modes 5 and 6, only the flux from the center legs flows through the transformer. Therefore, the transformer formed by the center leg operates independently from the transformer in the outer leg, and the equivalent magnetic circuit for this case is shown in Fig. 7(b).

As derived in the expanded time-variant equations for Mode 2 and Mode 5, the linear switching ramps are continuously bounded by the low-frequency cycle of v_{in}. Consequently, both the instantaneous switch current stress and the peak core flux density vary dynamically across the line period. The converter

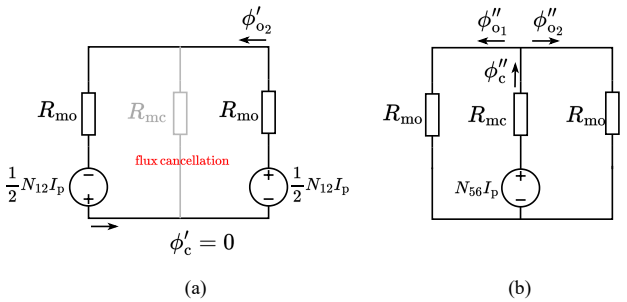


Fig. 7. Equivalent magnetic circuits for the proposed transformer. (a) shows the equivalent magnetic circuit for $N_{12}-N_{34}$, while (b) shows the equivalent magnetic circuit for $N_{56}-N_{78}$.

components experience their absolute maximum operational stresses exclusively at the peak of the AC line voltage $V_{\mathrm{in}_{\max}}$.

The proposed winding structure achieves zero magnetic coupling between the two transformers. Consequently, the distinct magnetic circuits share a single core without magnetic interference. Each transformer maintains independent operation despite the physical integration. This structure eliminates the need for two discrete magnetic components. As a result, the total volume of the magnetic components is significantly reduced.

Because the proposed integrated transformer achieves magnetic decoupling, the fundamental DCM operation of the converter remains electrically identical to a conventional interleaved topology utilizing discrete transformers. As a result, the standard electrical benefits of two-phase interleaving are fully preserved.

Furthermore, operating the rectifier in DCM with a fixed f_{s} and constant D, inherently achieves PFC because the average input current $\langle i_{\mathrm{in}}(t)\rangle$ tracks the $v_{\mathrm{in}}(t)$. This inherent behavior is derived as follows.

Using (2), the average input current $\langle i_{\mathrm{S}_1}(t)\rangle$ for the first phase over one T_{s} is given by (6):

$$\begin{aligned}\langle i_{\mathrm{S}_1}(t)\rangle &= \frac{1}{T_{\mathrm{s}}}\int_0^{T_{\mathrm{s}}} i_{\mathrm{S}_1}(\tau)d\tau \\ &= \frac{1}{2} i_{\mathrm{S}_{1_{\mathrm{pk}}}}(t)D = \frac{v_{\mathrm{in}}(t)D^2T_{\mathrm{s}}}{2L_{\mathrm{m}_1}}\end{aligned} \tag{6}$$

Because the interleaved phases are symmetrical and share the same magnetizing inductance with $L_{\mathrm{m}_1} = L_{\mathrm{m}_2} = L_{\mathrm{m}}$, the second phase draws an identical average current $\langle i_{\mathrm{S}_2}(t)\rangle$. The total average input current $\langle i_{\mathrm{in}}(t)\rangle$, is the sum of both active phases:

$$\langle i_{\mathrm{in}}(t)\rangle = \langle i_{\mathrm{S}_1}(t)\rangle + \langle i_{\mathrm{S}_2}(t)\rangle = \frac{D^2T_{\mathrm{s}}}{L_{\mathrm{m}}} v_{\mathrm{in}}(t) \tag{7}$$

Since D, L_{m}, and f_{s} are constant, $\langle i_{\mathrm{in}}(t)\rangle$ is directly proportional to $v_{\mathrm{in}}(t)$. Consequently, $i_{\mathrm{in}}(t)$ inherently tracks the AC envelope, thus naturally achieving PFC. Because this fundamental electrical behavior remains identical to a standard interleaved topology operating in DCM, both the PFC and Total Harmonic Distortion (THD) performance are expected to match that of a conventional discrete setup.

B. Saturation Magnetic Flux Density Analysis

The circuit operation is classified into three patterns determined by D and the boundary condition $D \leq 0.5\frac{nV_{\mathrm{o}}}{V_{\mathrm{in}_{\max}}}$. This classification, illustrated in Fig. 8, arises because the integrated single-core structure permits simultaneous magnetizing current flow in multiple active windings, resulting in distinct flux superposition compared to conventional discrete transformers. The manner of superposition due to the magnetizing current can be controlled by t_{dis}. Consequently, the saturation magnetic flux density $B_{\max}$ is a critical parameter for the transformer geometry. It can be expressed in terms of the saturation magnetic flux density in each phase, denoted as $B'_{\max}$ and $B''_{\max}$, with $B'_{\max} = B''_{\max}$ since $L_{\mathrm{m}_1} = L_{\mathrm{m}_2}$. The flux density in each leg is examined further using the principle of superposition, with the expressions of the superimposed flux densities in the left, right and center legs shown in (8)-(10), represented as B_{o_1}, B_{o_2} and B_{c} respectively:

$$B_{\mathrm{o}_1} = B'_{\mathrm{o}_1} + B''_{\mathrm{o}_1} \tag{8}$$

$$B_{\mathrm{o}_2} = B'_{\mathrm{o}_2} - B''_{\mathrm{o}_2} \tag{9}$$

$$B_{\mathrm{c}} = B''_{\mathrm{c}} \tag{10}$$

Case 1 (Fig. 8(a)), $D \leq 0.5\frac{nV_{\mathrm{o}}}{V_{\mathrm{in}_{\max}}}$*:* The waveforms in Fig. 4 represent the circuit operation for Case 1. As shown, flux cancellation does not occur in the outer legs; however, the magnitude of the peak flux density remains uniform across all legs. If the magnetizing current of one phase reaches zero before the magnetizing current of the other phase reaches its peak, then $B_{\max} = B'_{\max}$. This condition is satisfied when D is expressed by (11).

$$D \leq 0.5\frac{nV_{\mathrm{o}}}{V_{\mathrm{in}_{\max}}} \tag{11}$$

Case 2 (Fig. 8(b)), $0.5\frac{nV_{\mathrm{o}}}{V_{\mathrm{in}_{\max}}} < D \leq 0.5$*:* In this case, the magnetizing current of one phase does not reach zero before the subsequent phase reaches its peak. The saturation flux density across all phases can be derived as follows. At t_2, $B'_{\mathrm{o}_1} = B'_{\max}$. Moreover, $B''_{\mathrm{o}_1}(t_2) = B'_{\mathrm{o}_1}(t_5)$ and is expressed in (12):

$$B''_{\mathrm{o}_1}(t_2) = B'_{\mathrm{o}_1}(t_5) = B'_{\max} - \frac{B'_{\max}}{t_6 - t_2}(t_5 - t_2) \tag{12}$$

with $t_5 = 0.5T_{\mathrm{s}} + t_2$ and $t_2 = DT_{\mathrm{s}}$. The peak $i_{L_{\mathrm{m}_{1_{\mathrm{pk}}}}}$ can be rewritten as in (13):

$$i_{L_{\mathrm{m}_{1_{\mathrm{pk}}}}} = \frac{V_{\mathrm{in}_{\max}}}{L_{\mathrm{m}}} DT_{\mathrm{s}} = \frac{nV_{\mathrm{o}}}{L_{\mathrm{m}}}(t_6 - t_2) \tag{13}$$

In this case, (11) is not satisfied, and this results in $B_{\max}$ in the left outer leg exceeding $B'_{\max}$ in the center leg, as expressed in (14):

$$B_{\max} = \left(2 - \frac{0.5nV_{\mathrm{o}}}{V_{\mathrm{in}_{\max}} D}\right) B'_{\max} \tag{14}$$

Case 3 (Fig. 8(c)), $0.5 < D$*:* In this case, $B_{\max}$ in the left outer leg consistently exceeds $B'_{\max}$. By analysis similar to Case 2, $B_{\max}$ is derived as (15):

$$B_{\max} = \left(1 + \frac{D - 0.5}{D}\right) B'_{\max} \tag{15}$$

Because the flyback is interleaved, it is expected that the rectifier will primarily operate at $D \leq 0.5$, which operates in the conditions set by Case 1 or 2.

C. Design Method of the Proposed Integrated Transformer

A turns ratio n is selected based on the maximum voltage ratings of the switches. The maximum operating duty cycle D is selected in consideration of the desired operation discussed in Section II-B. The magnetizing inductance, $L_{\mathrm{m}_1} = L_{\mathrm{m}_2} = L_{\mathrm{m}}$ is calculated from the circuit specifications. The saturation magnetic flux density per phase $B'_{\max}$ is calculated from the $B_{\max}$ given by the chosen core material, considering the maximum D. A core is chosen using the AP method.

The number of turns is calculated using $B'_{\max}$ and the peak primary side current $I_{\mathrm{pri}_{\mathrm{pk}}}$ as shown in (16) and (17):

$$\frac{N_{12}}{n} = N_{34} \geq \frac{L_{\mathrm{m}} I_{\mathrm{pri}_{\mathrm{pk}}}}{n A_{\mathrm{mo}} B'_{\max}} \tag{16}$$

$$\frac{N_{56}}{n} = N_{78} \geq \frac{L_{\mathrm{m}} I_{\mathrm{pri}_{\mathrm{pk}}}}{n A_{\mathrm{mc}} B'_{\max}} \tag{17}$$

The magnetic resistances of the outer legs R_{mo} and the center leg R_{mc} are obtained from a magnetic circuit analysis of the integrated transformer in its separate phases. The transformer $N_{12}-N_{34}$ is only made of the outer legs as shown in Fig. 9(a), while $N_{56}-N_{78}$ is made of all the legs. The magnetizing inductance L_{m} in terms of the number of turns and magnetic resistances for $N_{12}-N_{34}$ and $N_{56}-N_{78}$ are shown in (18) and (19) respectively.

$$L_{\mathrm{m}} = \frac{N_{12}^2}{2R_{\mathrm{mo}}} \tag{18}$$

$$L_{\mathrm{m}} = \frac{N_{56}^2}{R_{\mathrm{mc}} + 0.5R_{\mathrm{mo}}} \tag{19}$$

The magnetic resistances R_{mo} and R_{mc} are obtained from (20) and (21):

$$R_{\mathrm{mo}} = \frac{N_{12}^2}{2L_{\mathrm{m}}} \tag{20}$$

$$R_{\mathrm{mc}} = \frac{(2N_{56} - N_{12})(2N_{56} + N_{12})}{4L_{\mathrm{m}}} \tag{21}$$

Finally, an air gap was also inserted in each of the legs to provide the desired resistance. The magnetic path lengths, l_{mo} and l_{mc}, and cross-sectional areas, A_{mo} and A_{mc}, of the outer and center legs respectively are shown in Fig. 9(b). Non-equal length air gaps were placed in the transformer. There are two cases, one where the air gaps in the outer legs $l_{\mathrm{mo}_{\mathrm{air}}}$ are longer than the air gap in the center leg $l_{\mathrm{mc}_{\mathrm{air}}}$, and one where $l_{\mathrm{mc}_{\mathrm{air}}}$ is longer than $l_{\mathrm{mo}_{\mathrm{air}}}$.

Case 1: $l_{\mathrm{mo}_{\mathrm{air}}} > l_{\mathrm{mc}_{\mathrm{air}}}$: Given the setup in Fig. 10(a), R_{mo} and R_{mc} are shown in (22) and (23) respectively with $l_{\mathrm{mo}_{\mathrm{air}}}$ and $l_{\mathrm{mc}_{\mathrm{air}}}$ in (24) and (25) respectively.

$$R_{\mathrm{mo}} = \frac{l_{\mathrm{mo}} - l_{\mathrm{mo}_{\mathrm{air}}} + l_{\mathrm{mc}_{\mathrm{air}}}}{\mu_r \mu_0 A_{\mathrm{mo}}} + \frac{l_{\mathrm{mo}_{\mathrm{air}}}}{\mu_0 A_{\mathrm{mo}}} \tag{22}$$

$$R_{\mathrm{mc}} = \frac{l_{\mathrm{mc}}}{\mu_r \mu_0 A_{\mathrm{mc}}} + \frac{l_{\mathrm{mc}_{\mathrm{air}}}}{\mu_0 A_{\mathrm{mc}}} \tag{23}$$

$$l_{\mathrm{mo}_{\mathrm{air}}} = \frac{\mu_0(\mu_r A_{\mathrm{mo}} R_{\mathrm{mo}} - A_{\mathrm{mc}} R_{\mathrm{mc}}) + \frac{l_{\mathrm{mc}}}{\mu_r} - l_{\mathrm{mo}}}{\mu_r - 1} \tag{24}$$

$$l_{\mathrm{mc}_{\mathrm{air}}} = \mu_0 A_{\mathrm{mc}} R_{\mathrm{mc}} - \frac{l_{\mathrm{mc}}}{\mu_r} \tag{25}$$

Case 2: $l_{\mathrm{mo}_{\mathrm{air}}} < l_{\mathrm{mc}_{\mathrm{air}}}$: Given the setup in Fig. 10(b), R_{mo} and R_{mc} are shown in (26) and (27) respectively, $l_{\mathrm{mo}_{\mathrm{air}}}$ and $l_{\mathrm{mc}_{\mathrm{air}}}$ in (28) and (29) respectively.

$$R_{\mathrm{mo}} = \frac{l_{\mathrm{mo}}}{\mu_r \mu_0 A_{\mathrm{mo}}} + \frac{l_{\mathrm{mo}_{\mathrm{air}}}}{\mu_0 A_{\mathrm{mo}}} \tag{26}$$

$$R_{\mathrm{mc}} = \frac{l_{\mathrm{mc}} - l_{\mathrm{mc}_{\mathrm{air}}} + l_{\mathrm{mo}_{\mathrm{air}}}}{\mu_r \mu_0 A_{\mathrm{mc}}} + \frac{l_{\mathrm{mc}_{\mathrm{air}}}}{\mu_0 A_{\mathrm{mc}}} \tag{27}$$

$$l_{\mathrm{mo}_{\mathrm{air}}} = \mu_0 A_{\mathrm{mo}} R_{\mathrm{mo}} - \frac{l_{\mathrm{mo}}}{\mu_r} \tag{28}$$

$$l_{\mathrm{mc}_{\mathrm{air}}} = \frac{\mu_0(\mu_r A_{\mathrm{mc}} R_{\mathrm{mc}} - A_{\mathrm{mo}} R_{\mathrm{mo}}) + \frac{l_{\mathrm{mo}}}{\mu_r} - l_{\mathrm{mc}}}{\mu_r - 1} \tag{29}$$

III. Simulation Verification of Proposed Design

A. Simulation Verification of the Proposed Design Method

To verify the operation of the integrated transformer, a simulation of the two-phase interleaved flyback rectifier was first performed in PLECS Standalone 4.9.4, with the diagram shown in Fig. 11(a). Clamped RCD Snubbers ($C_{\mathrm{sn}_1}, D_{\mathrm{sn}_1}, R_{\mathrm{sn}_1}$ and $C_{\mathrm{sn}_2}, D_{\mathrm{sn}_2}, R_{\mathrm{sn}_2}$) were also added for protection against surge voltage [37], [38] caused by the

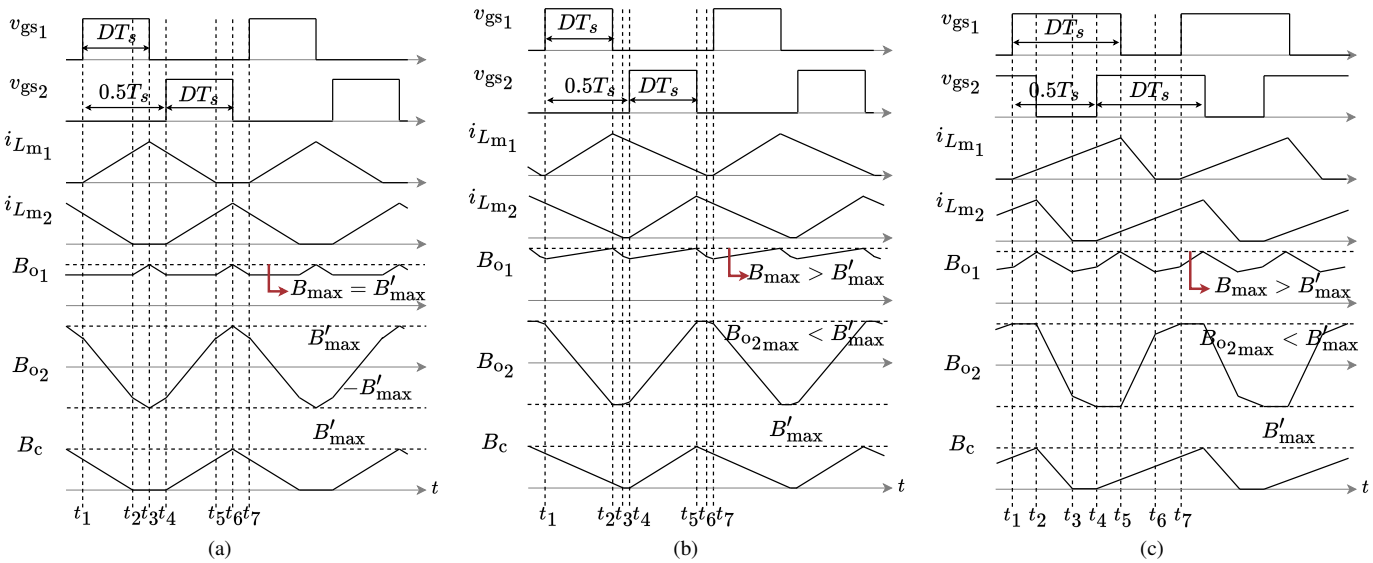
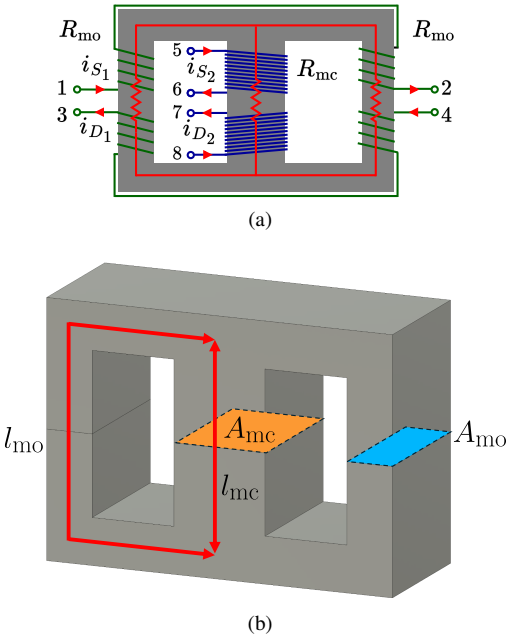


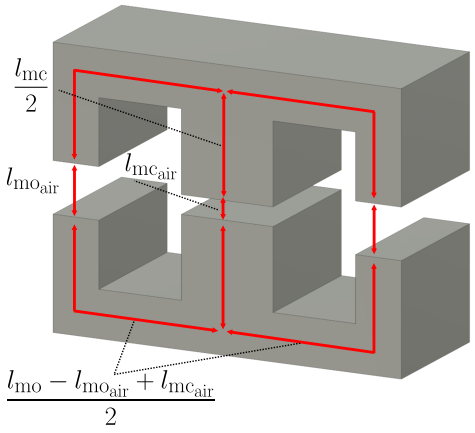
Fig. 8. Theoretical, superimposed magnetic flux density waveforms in the integrated transformer during circuit operations: (a) Case 1, (b) Case 2, and (c) Case 3.



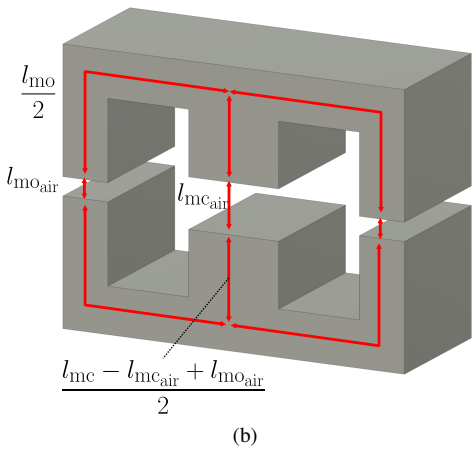
(a)

(b)

Fig. 9. EE core configuration: (a) magnetic resistances of the transformer (R_{mo} for the outer legs, R_{mc} for the center leg), and (b) effective cross-sectional area (A_{mo} for the outer legs, A_{mc} for the center leg) and effective magnetic path length l_{mo} for the outer legs, l_{mc} for the center leg.

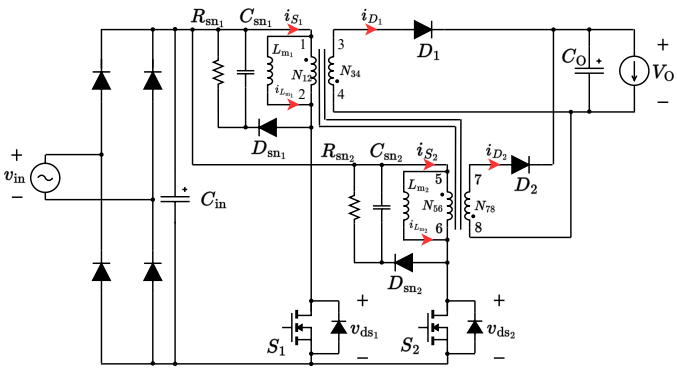


(a)

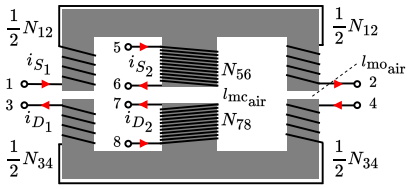


(b)

Fig. 10. EE core configuration with air gaps: (a) setup when $l_{\mathrm{mo_{air}}} > l_{\mathrm{mc_{air}}}$ and (b) setup when $l_{\mathrm{mo_{air}}} < l_{\mathrm{mc_{air}}}$.



(a)



(b)

Fig. 11. Schematic of the: (a) prototype two-phase interleaved flyback rectifier and (b) proposed integrated transformer.

leakage inductance and the switch capacitance at the turn-off of S_1 and S_2. An integrated transformer, shown in Fig. 10(b), was also designed using the proposed method. The circuit and transformer were designed with the condition given by (11), and will thus be verified in this study. The specifications of the two-phase interleaved flyback rectifier and transformer for the simulation verification are shown in Tables I and II respectively. The duty cycle is calculated using (30) [6].

$$D = \frac{2}{V_{\mathrm{in_{pk}}}} \sqrt{\frac{0.5 L_{\mathrm{m}} P_{\mathrm{o}}}{T_{\mathrm{s}}}} \tag{30}$$

B. Simulation Results

Fig. 12 shows waveforms from the simulation testing of the two-phase interleaved flyback rectifier with the proposed transformer at 240 W. Key waveforms of the circuit were observed from the simulation, which include $v_{\mathrm{gs_1}}$ and $v_{\mathrm{gs_2}}$, $i_{L_{\mathrm{m_1}}}$ and $i_{L_{\mathrm{m_2}}}$, and the fluxes $B_{\mathrm{o_1}}$, $B_{\mathrm{o_2}}$, and B_{c}. Similar waveforms to that of Fig. 4 have been observed, confirming that the system operates exactly as expected from the theoretical analysis. Flux cancellation is observed in the center

TABLE I
SIMULATION SPECIFICATIONS OF THE PROTOTYPE TWO-PHASE INTERLEAVED FLYBACK RECTIFIER

Item	Symbol	Value
AC input voltage	v_{in}	100 V_{rms}
Maximum rated power	P_{O}	240 W
Output voltage	V_{O}	48 V
Switching frequency	f_{s}	100 kHz
Input capacitor	C_{in}	1 µF
Output capacitor	C_{O}	2 x 1mF (parallel)
Snubber capacitor	$C_{\mathrm{sn_1}}, C_{\mathrm{sn_2}}$	2.2 nF
Snubber resistor	$R_{\mathrm{sn_1}}, R_{\mathrm{sn_2}}$	30 kΩ

TABLE II
SIMULATION SPECIFICATIONS OF THE PROTOTYPE INTEGRATED TRANSFORMER

Item	Symbol	Value
Magnetic core	-	OP44721EC
Relative permeability	μ_r	2500
Magnetic path length (center leg)	l_{mc}	31.7 mm
Magnetic path length (outer leg)	l_{mo}	71.4 mm
Cross-sectional area (center leg)	A_{mc}	243.4 mm^2
Cross-sectional area (outer leg)	A_{mo}	117.6 mm^2
Magnetizing inductance	$L_{\mathrm{m}_1}, L_{\mathrm{m}_2}$	82.8 µH
Turn ratio	n	3
Number of turns	$N_{12} : N_{34}$ $N_{56} : N_{78}$	24:8 18:6
Center leg air gap	$l_{\mathrm{mc}_{\mathrm{air}}}$	0.650 mm
Outer leg air gap	$l_{\mathrm{mo}_{\mathrm{air}}}$	0.500 mm
Coupling coefficient of N_{12}, N_{34}	$k_{N_{12}-N_{34}}$	0.990
Coupling coefficient of N_{56}, N_{78}	$k_{N_{56}-N_{78}}$	0.990
Coupling coefficient of N_{12}, N_{56}	$k_{N_{12}-N_{56}}$	0.000

leg during the time that $N_{56} - N_{78}$ is inactive, as denoted by the interval when $i_{L_{\mathrm{m}_2}} = 0$. The magnetizing currents $i_{L_{\mathrm{m}_1}}$ and $i_{L_{\mathrm{m}_2}}$ subside to zero prior to the subsequent phase reaching its peak. Moreover, the increase and decrease of fluxes were all linear, therefore confirming that there is indeed no magnetic coupling interference. This confirms the validity of the theoretical operation proposed in Section II.

IV. SIZE REDUCTION EFFECT OF THE PROPOSED TRANSFORMER

This section examines the size reduction effect by the integrated transformer. For this study, the conventional transformer is assumed to be two separate transformers that facilitate isolation between the primary and secondary side. The conventional transformer and its windings is represented in Fig. 13(a). The proposed integrated transformer and its windings is represented in Fig. 13(b). The volume is compared by using the AP method. The AP method utilizes the product between the core cross-sectional area A_{c}, and the window area A_{w}, and is a fundamental core size for the application [39]. It can be expressed both in terms of the transformer geometric parameters and the circuit parameters, as will be demonstrated in the following analysis.

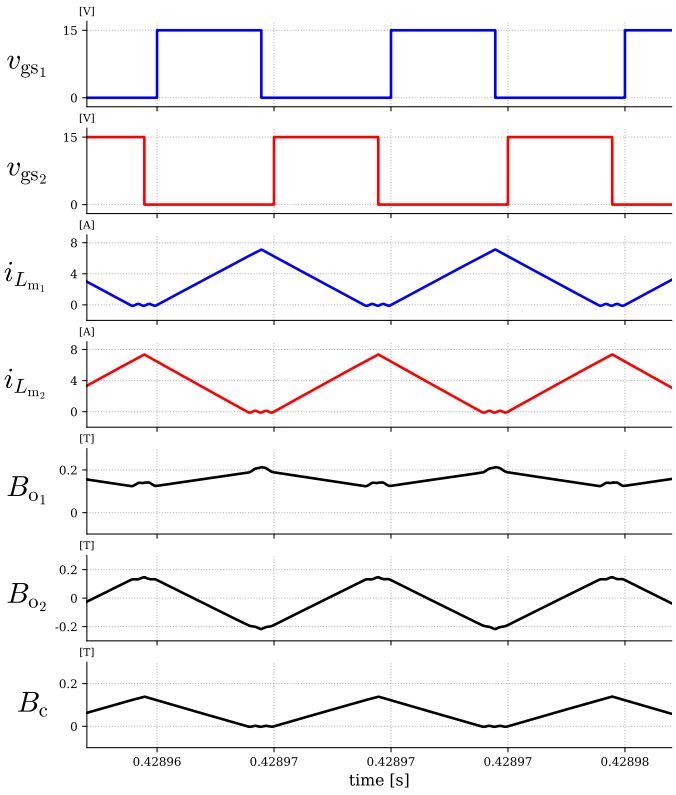


Fig. 12. Key simulation waveforms of the proposed two-phase interleaved flyback rectifier at 240 W.

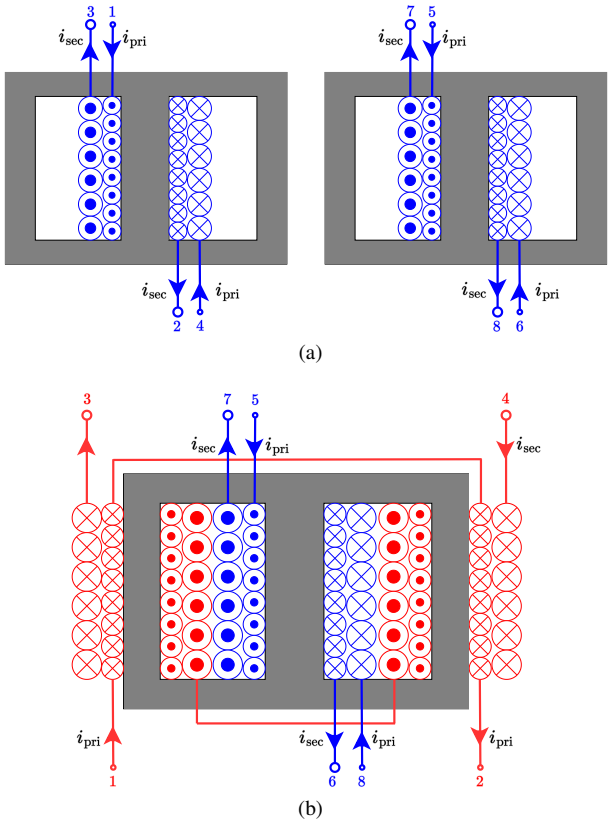


Fig. 13. Two separate transformers (a) and the proposed integrated transformer (b) used in the volume analysis, with the windings represented as well.

The EE-core was redesigned such that A_{w} and A_{c} are squares to allow for simpler analysis, as shown in Fig. 14 [40]. Here, A_{c} is twice the area of the outer leg.

For the conventional transformer, A_{c} is expressed in (31). Here, N_{p} is the number of turns wound on the center leg of the conventional transformer.

$$A_{\mathrm{c}_{\mathrm{conv}}} = \frac{L_{\mathrm{m}} I_{\mathrm{pri}_{\mathrm{pk}}}}{B_{\mathrm{max}} N_{\mathrm{p}}} \tag{31}$$

For the integrated transformer, A_{c} is expressed in (32).

$$A_{\mathrm{c}_{\mathrm{integ}}} = \frac{L_{\mathrm{m}} I_{\mathrm{pri}_{\mathrm{pk}}}}{B_{\mathrm{max}} N_{56}} \tag{32}$$

The window area A_{w} can be expressed in terms of winding area and the occupancy ratio K_{u}, as seen in (33) for the conventional transformer:

$$A_{\mathrm{w}_{\mathrm{conv}}} K_{\mathrm{u}} = N_{\mathrm{p}} \frac{I_{\mathrm{pri}_{\mathrm{rms}}}}{J_{\mathrm{m}}} + N_{\mathrm{s}} \frac{I_{\mathrm{sec}_{\mathrm{rms}}}}{J_{\mathrm{m}}} \tag{33}$$

which simplifies to:

$$A_{\mathrm{w_{conv}}} = \frac{N_{\mathrm{p}}}{J_{\mathrm{m}} K_{\mathrm{u}}} \left(I_{\mathrm{pri_{rms}}} + \frac{I_{\mathrm{sec_{rms}}}}{n} \right) \tag{34}$$

For the integrated transformer, winding N_{12} and N_{34} on the outer legs implies that $N_{12} \approx 2N_{56}$ and $N_{34} \approx 2N_{78}$. Taking into account that the core window has half the total turns of the transformer $N_{12} - N_{34}$, A_{w} can be expressed as (35):

$$\begin{aligned} A_{\mathrm{w_{integ}}} K_{\mathrm{u}} &= \left(N_{56} \frac{I_{\mathrm{pri_{rms}}}}{J_{\mathrm{m}}} + N_{78} \frac{I_{\mathrm{sec_{rms}}}}{J_{\mathrm{m}}} \right) \\ &\quad + \frac{1}{2} \left(N_{12} \frac{I_{\mathrm{pri_{rms}}}}{J_{\mathrm{m}}} + N_{34} \frac{I_{\mathrm{sec_{rms}}}}{J_{\mathrm{m}}} \right) \\ &= 2 \left(\frac{N_{56} I_{\mathrm{pri_{rms}}}}{J_{\mathrm{m}}} + \frac{N_{78} I_{\mathrm{sec_{rms}}}}{J_{\mathrm{m}}} \right) \end{aligned} \tag{35}$$

Using (31), (32), (34), and (35), the AP of the conventional and proposed integrated transformers are expressed in (36) and (37) respectively:

$$\begin{aligned} AP_{\mathrm{conv}} &= A_{\mathrm{c_{conv}}} A_{\mathrm{w_{conv}}} \\ &= \frac{L_{\mathrm{m}} I_{\mathrm{pri_{pk}}} \left(I_{\mathrm{pri_{rms}}} + \frac{I_{\mathrm{sec_{rms}}}}{n} \right)}{B_{\max} J_{\mathrm{m}} K_{\mathrm{u}}} \end{aligned} \tag{36}$$

$$\begin{aligned} AP_{\mathrm{integ}} &= A_{\mathrm{c_{integ}}} A_{\mathrm{w_{integ}}} \\ &= \frac{2 L_{\mathrm{m}} I_{\mathrm{pri_{pk}}} \left(I_{\mathrm{pri_{rms}}} + \frac{I_{\mathrm{sec_{rms}}}}{n} \right)}{B_{\max} J_{\mathrm{m}} K_{\mathrm{u}}} \end{aligned} \tag{37}$$

From (36) and (37), $AP_{\mathrm{integ}} \approx 2AP_{\mathrm{conv}}$ ideally.

The total volume of the EE core in Fig. 14 is expressed in (38), where x is a constant that denotes $A_{\mathrm{c}}/A_{\mathrm{w}}$. The volume corresponding to the conventional and integrated transformer is obtained using their respective A_{c} and A_{w}. Since the A_{c} and A_{w} depend on the number of corresponding turns, then the estimated volume will change depending on the actual implementation of the transformer.

$$\mathrm{Vol} = 2AP^{\frac{3}{4}} \left(2x^{\frac{1}{4}} + x^{\frac{3}{4}} + x^{-\frac{1}{4}} \right) \tag{38}$$

Also, the A_{c} and A_{w} can be expressed in terms of the AP and x as shown in (39) and (40):

$$A_{\mathrm{c}} = \sqrt{x \cdot AP} \tag{39}$$

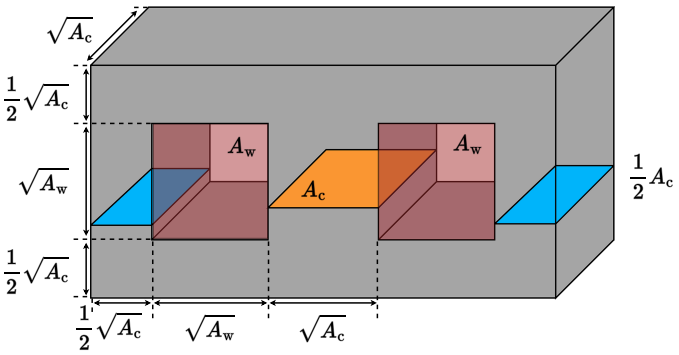


Fig. 14. Simplified representation of one EE-core used in the volume analysis.

$$A_{\mathrm{w}} = \sqrt{\frac{AP}{x}} \tag{40}$$

The rms expressions of the primary and secondary side current of the interleaved flyback rectifier are given in (41) and (42) [6].

$$I_{\mathrm{pri_{rms}}} = \frac{V_{\mathrm{in_{max}}}}{L_{\mathrm{m}} f_{\mathrm{s}}} \sqrt{\frac{D^3}{6}} \tag{41}$$

$$I_{\mathrm{sec_{rms}}} = \frac{V_{\mathrm{in_{max}}}}{L_{\mathrm{m}} f_{\mathrm{s}}} \sqrt{\frac{4nD^3}{9\pi} \cdot \frac{V_{\mathrm{in_{max}}}}{V_{\mathrm{o}}}} \tag{42}$$

The AP calculation based on circuit specifications is summarized in Table III. The conventional transformer implementation comprises of two discrete cores, the value indicated in this table is for one core only.

The theoretical volume is dominated by the $AP^{3/4}$ scaling law as seen in (38), and thus the term $\left(2x^{1/4} + x^{3/4} + x^{-1/4}\right)$ can be treated as constant. Using the relationship $AP_{\mathrm{integ}} \approx 2AP_{\mathrm{conv}}$, then the amount of volume reduction can be computed using (38), with the reduction amounting to 16%. This assumes that the integrated core geometry is custom-tailored to the specific AP requirements derived from the circuit specifications, rather than being constrained by standard commercial core shapes. While the experimental prototype described in Section V utilizes a commercial core (Magnetics 0P44721EC) for proof-of-concept, this theoretical analysis confirms that a purpose-built integrated core would offer significant improvements in power density compared to the conventional discrete implementation by standard commercial core shapes.

While the proposed integrated transformer achieves a theoretical 16% reduction in core volume, this structural integration introduces some trade-offs regarding core and winding losses.

In a conventional discrete setup, the flux density remains uniform across the isolated magnetic paths. In contrast, the proposed integrated core shares flux paths, resulting in a complex flux distribution due to the superposition of individual phase contributions. As established in the theoretical waveforms of Fig. 8, overlapping conduction intervals create localized regions of high flux density (additive flux) in one

TABLE III
AREA PRODUCTS OF THE CONVENTIONAL (1 CORE) AND PROPOSED INTEGRATED TRANSFORMER

Item	Symbol	Conventional (1 core)	Proposed
Maximum input voltage	$V_{\mathrm{in_{max}}}$	240 V_{rms}	
Output voltage	V_{O}	48 V	
Maximum duty cycle	D	0.47	
Switching frequency	f_{s}	100 kHz	
Magnetizing inductance	L_{m}	82.8 µH	
Turn ratio	n	3	
Primary side peak current	$I_{\mathrm{pri,pk}}$	8.02 A	
Primary side rms current	$I_{\mathrm{pri,rms}}$	5.39 A	
Secondary side rms current	$I_{\mathrm{sec,rms}}$	12.53 A	
Saturation flux density	$B_{\max}$	0.3 T	
Maximum current density	J_{m}	5 A/mm^2	
Occupancy ratio	K_{u}	0.3	
Area Product	AP	14128.75 mm^4	28257.50 mm^4

outer leg, while simultaneously creating regions of reduced flux density (subtractive flux) in the opposite outer leg. Because core loss scales non-linearly with flux density, the net impact on the total core loss involves both mitigating and compounding factors, necessitating rigorous analysis for full volumetric loss integration in future optimization stages.

Furthermore, the geometry of the shared EE-core forces a structural asymmetry in the windings. To balance the magnetizing inductance, the outer legs require an increased turn count compared to a conventional discrete transformer. Although the smaller physical perimeter of the outer legs partially offsets the DC resistance penalty of these additional turns, the high-frequency AC resistance R_{ac} remains highly dependent on the physical layering and winding arrangement. Consequently, achieving optimal winding efficiency requires careful geometric optimization and parameter sweeping, which is a critical focus for the future phases of this study.

V. Experimental Verification of Proposed Design

A. Prototype Flyback Circuit and Integrated Transformer

A prototype two-phase interleaved flyback rectifier, with the circuit schematic shown in Fig. 11(a), was built to verify the operation of the integrated transformer operating in DCM. A photograph of the prototype evaluation board is provided in Fig. 15, with the primary and secondary side separated for simple implementation. Because the fundamental operating principles of the converter remain identical to a conventional discrete setup, standard closed-loop modulation methods are directly applicable. Therefore, the experimental evaluation was intentionally conducted using open-loop control with a fixed duty cycle decided using (30).

The prototype integrated transformer, represented in Fig. 11(b), was also designed using the proposed method. To strictly minimize leakage inductance, the primary and secondary windings of the transformer for each respective phase were physically overlapped on their designated legs. The EE-core used for the prototype is shown in Fig. 16(a). The windings are wound around lab-made coil formers, as shown in Fig. 16(b), which were fabricated using a Flashforge Adventurer 4 Series 3D printer, and are finally placed onto the corresponding legs of the core. A photograph of the fully assembled prototype integrated transformer is shown in Fig. 16(c). The specifications of the two-phase interleaved flyback rectifier and transformer are shown in Tables IV and V, respectively.

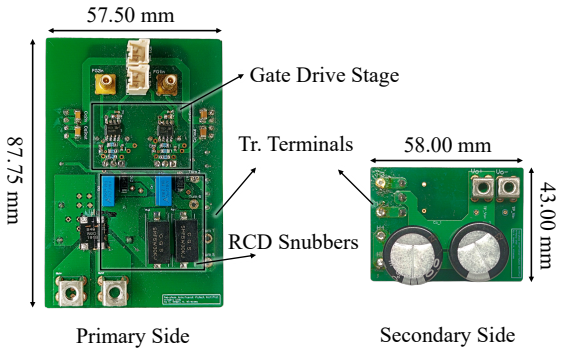


Fig. 15. Photograph of the evaluation board for the proposed integreated flyback PFC rectifier.

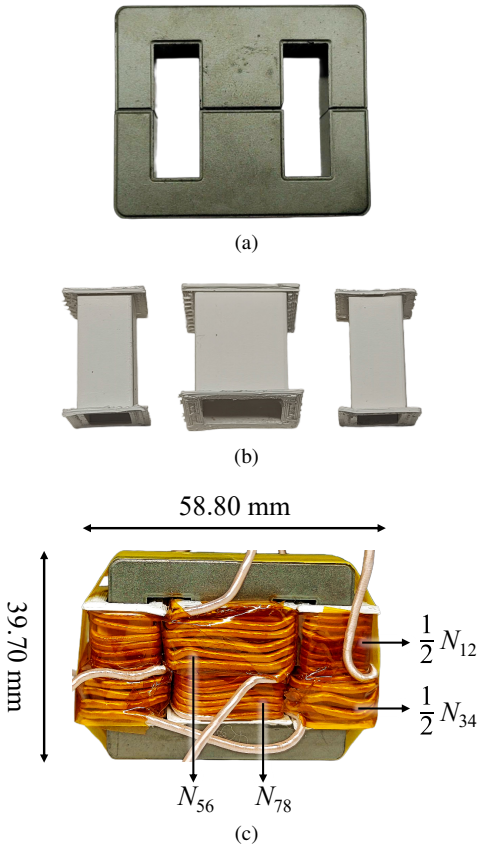


Fig. 16. Photographs of the proposed integrated transformer: (a) EE-core for the prototype; (b) lab-produced coil formers using a Flashforge Adventurer 4 Series 3D printer; and (c) final assembled prototype, with windings labeled.

The comprehensive list of equipment used in the experimental evaluation is summarized in Table VI. A Tektronix MSO58 high-definition oscilloscope was used to capture experimental waveforms. The open-loop PWM signals were generated via a Tektronix AFG31000 Series function generator, and the integrated transformer parameters were characterized using a Hioki IM3536 LCR meter at 100 kHz.

The coupling coefficients of the transformers $k_{N_{12}-N_{34}}$, $k_{N_{56}-N_{78}}$, and $k_{N_{12}-N_{56}}$ were derived from the mutual inductance obtained after connecting the respective transformer windings in series-aiding and series-opposing configurations. The physical overlapping of the windings resulted in high intra-phase coupling coefficients, with $k_{N_{12}-N_{34}}$ measured at 0.970 and $k_{N_{56}-N_{78}}$ at 0.979. Furthermore, the cross-coupling coefficient between the two flyback transformers $k_{N_{12}-N_{56}}$ is 0.0005, showing that the coupling between phases was almost eliminated. Because this decoupling is geometrically enforced by the core and winding arrangement, it is expected to be maintained even under load transient conditions. While severe transients may introduce high-frequency harmonics that slightly degrade this ideal isolation due to parasitic elements, their magnitude and energy are negligible compared to the fundamental operating frequencies, ensuring that the overall dynamic stability of the converter is preserved.

B. Experimental Results

Fig. 17(a) shows the 100 W experimental waveforms at 4 ms/div from the two-phase interleaved flyback rectifier with the proposed transformer, demonstrating PFC. Fig. 17(b) presents 2 µs/div waveforms. No current is induced in either switch except during the resonance between the leakage inductance and the parasitic capacitance of the switch. During the inactive period of the transformer, secondary diode side current is zero. Moreover, after i_{S_1} and i_{S_2} reached zero when S_1 and S_2 switched off, the currents remained at zero. Therefore, the two flyback converters operate independently without magnetic coupling interference. The same phenomenon was observed at 240 W two-phase drive, as shown in Fig. 17(c). Single-phase drive at lower power at 32.5 W is also feasible as shown in Fig. 17(d).

Two distinct parasitic resonant behaviors can be observed in the v_{ds} waveforms in Fig. 17(b) and Fig. 17(c) according

TABLE IV
SPECIFICATIONS OF THE PROTOTYPE TWO-PHASE INTERLEAVED FLYBACK RECTIFIER

Item	Symbol	Value / Component
AC input voltage	v_{in}	100-240 V_{rms}
Maximum rated power	P_{O}	240 W
Output voltage	V_{O}	48 V
Switching frequency	f_{s}	100 kHz
Bridge diode	BD	CD-DF408S
Input capacitor	C_{in}	1 µF
Switch	S_1, S_2	STF15N80k5
Secondary diode	D_1, D_2	STPSC30H12CWL
Output capacitor	C_{O}	2 x 1mF, in parallel
Snubber capacitor	$C_{\mathrm{sn}_1}, C_{\mathrm{sn}_2}$	2.2 nF
Snubber resistor	$R_{\mathrm{sn}_1}, R_{\mathrm{sn}_2}$	30 kΩ
Snubber diode	$D_{\mathrm{sn}_1}, D_{\mathrm{sn}_2}$	VS-E7MH0112-M3/1

TABLE V
SPECIFICATIONS OF THE PROTOTYPE INTEGRATED TRANSFORMER

Item	Symbol	Value
Magnetizing inductance (Target)	L_{m}	82.8 µH
Magnetizing inductance of $N_{12} - N_{34}$	L_{m_1}	86.1 µH
Magnetizing inductance of $N_{56} - N_{78}$	L_{m_2}	75.1 µH
Turn ratio	n	3
Magnetic core	-	OP44721EC
Number of turns	$N_{12} : N_{34}$ $N_{56} : N_{78}$	24:8 15:5
Center leg air gap	$l_{\mathrm{mc}_{\mathrm{air}}}$	0.650 mm
Outer leg air gap	$l_{\mathrm{mo}_{\mathrm{air}}}$	0.500 mm
Coupling coefficient of N_{12}, N_{34}	$k_{N_{12}-N_{34}}$	0.970
Coupling coefficient of N_{56}, N_{78}	$k_{N_{56}-N_{78}}$	0.979
Coupling coefficient of N_{12}, N_{56}	$k_{N_{12}-N_{56}}$	0.0005

TABLE VI
EQUIPMENT USED IN THE EXPERIMENTAL VERIFICATION

Equipment Type	Manufacturer & Model
AC Power Supply	Kikusui CR2000M
DC Electronic Load	Kikusui PLZ1004WH
Oscilloscope (8-Channel)	Tektronix MSO58
Differential Voltage Probes	Tektronix THDP0200
Current Probes	Tektronix TCP0030A
Auxiliary Power Supply (5 V)	Matsusada TB160V15A720W
Auxiliary Power Supply (3.3 V)	Kikusui PWR401L
Function Generator	Tektronix AFG31000 Series
LCR Meter	Hioki IM3536

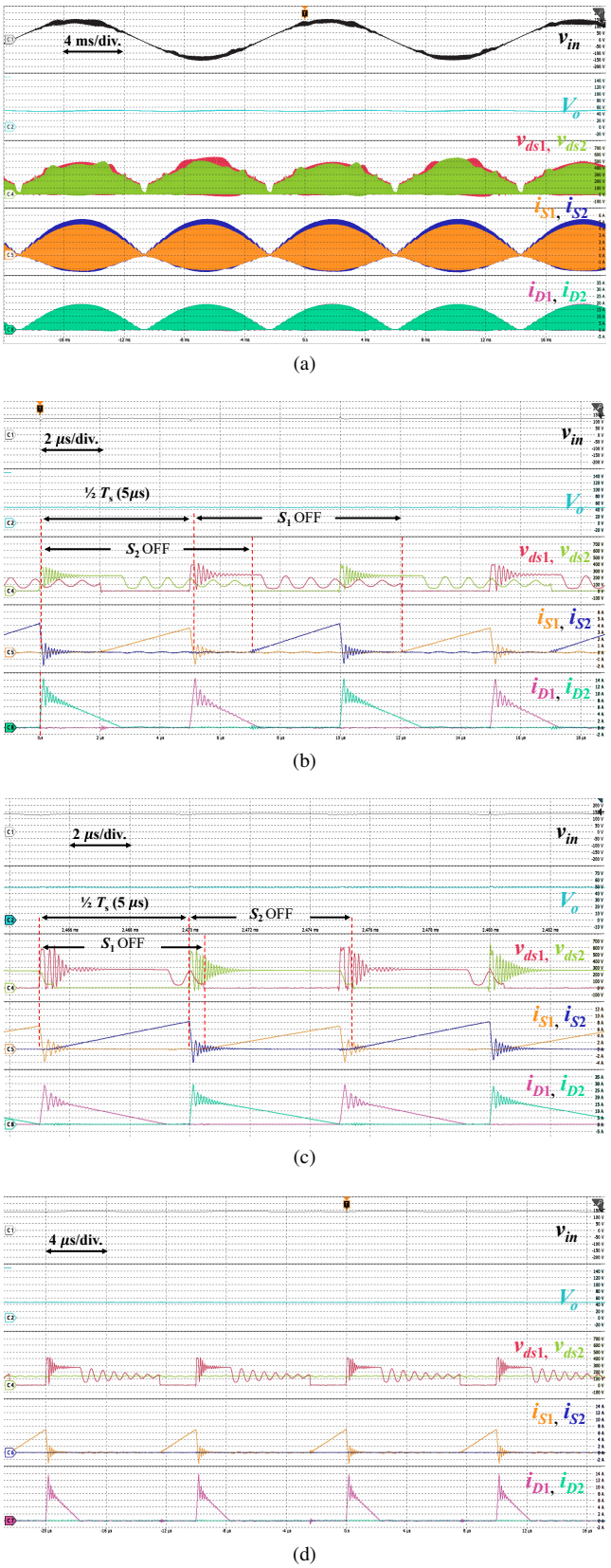


Fig. 17. Experimental waveforms of the proposed two-phase interleaved flyback rectifier at (a) & (b): 100 W, two-phase AC drive, (c): 240 W, two-phase AC drive, (d): 32.5 W, single-phase AC drive. For (a) and (b): v_{in}: 50 V/div., V_{o}: 10 V/div., $v_{\mathrm{ds}_1}, v_{\mathrm{ds}_2}$: 100 V/div., $i_{\mathrm{S}_1}, i_{\mathrm{S}_2}$: 1 A/div., $i_{\mathrm{D}_1}, i_{\mathrm{D}_2}$: 5 A/div.; For (c): v_{in}: 50 V/div., V_{o}: 10 V/div., $v_{\mathrm{ds1}}, v_{\mathrm{ds2}}$: 100 V/div., $i_{\mathrm{S}_1}, i_{\mathrm{S}_2}$: 2 A/div., $i_{\mathrm{D}_1}, i_{\mathrm{D}_2}$: 5 A/div.; For (d): v_{in}: 50 V/div., V_{o}: 20 V/div., $v_{\mathrm{ds}_1}, v_{\mathrm{ds}_2}$: 100 V/div., $i_{\mathrm{S}_1}, i_{\mathrm{S}_2}$: 2 A/div., $i_{\mathrm{D}_1}, i_{\mathrm{D}_2}$: 2 A/div.

to the operational dynamics of the flyback converter. The same phenomenon can also be observed in Fig. 17(d) for one-phase drive. A high-frequency ringing occurs immediately at switch turn-off, driven by the resonance between the leakage inductance of the transformer and the parasitic output capacitance C_{oss} of the switches. This voltage stress is mitigated by the implemented primary-side RCD snubbers. Second, a lower-frequency oscillation appears when the secondary diode currents i_{D_1} and i_{D_2} reach zero, indicating complete transformer demagnetization. During this interval, the main magnetizing inductance of the active transformer resonates with C_{oss} until the next switching cycle initiates. Nevertheless, these parasitic behaviors do not interfere with the non-coupled operation of the proposed transformer.

VI. Conclusion

In this paper, an integrated transformer was successfully designed for a two-phase interleaved flyback rectifier for USB PD applications. The proposed winding structure for the integrated transformer prevents coupling between the two flyback transformers and achieves simple manufacturability through the use of a commercial EE-core. A coupling coefficient of 0.0005 was achieved with the prototype. Independent flyback circuit operation without magnetic interference due to coupling was verified through simulation and experiment up to 240 W. Through the proposed design, two discrete transformers are integrated into a single magnetic component. The structure is estimated to enable up to a 16% downsizing of magnetic components compared to conventional discrete transformers.

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